

Optimizing I²C Slaver Architecture for Enhanced Efficiency in Low-Power Applications

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Abstract—This paper introduces an optimized I²C slave architecture designed to enhance efficiency in low-power applications, implemented and validated on FPGA. Traditional I²C slave designs often struggle with inefficient data handling and excessive power consumption, making them less suitable for modern embedded systems. To address these challenges, we propose a FIFO-based architecture with dynamic clock gating. This design efficiently manages variable data rates and reduces dynamic power consumption by enabling the clock only when necessary. Through FPGA-based implementation, we validate the functionality and power efficiency of the design, demonstrating improved performance, scalability, and reduced power usage compared to traditional approaches.

Keywords— I²C Slave, FIFO Buffer, Clock Gating, Low-Power Design, Verilog HDL, Serial Communication, FPGA Implementation.

I. INTRODUCTION

The Inter-Integrated circuit (I²C) slave architecture is designing and implemented using FPGA (Field-Programmable Gate Array) technology with the Xilinx software. The I²C protocol is widely used in modern digital systems due to its simplicity, low pin count, and ability to connect multiple devices efficiently over a two-wire communication bus. By leveraging FPGA technology, this implementation provides a flexible and high-performance solution that can be customized for various application. The increasing demand for compact, efficient, and low-power embedded systems has significantly influenced the evolution of communication protocols like I²C (Inter-Integrated Circuit). Originally developed by Philips, I²C has become an industry standard for enabling communication between multiple integrated circuits on the same board using just two lines: Serial Data (SDA) and Serial Clock (SCL). Its simplicity, low pin count, and robust support for multiple devices have made it a go-to choice for many embedded applications, ranging from sensor interfacing in mobile devices to complex communication in automotive control units. This research introduces an enhanced I²C slave architecture that addresses the inherent inefficiencies of conventional designs by incorporating two key techniques:

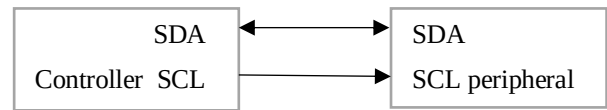
FIFO (First-In, First-Out) buffering and dynamic clock gating. The FIFO buffer allows the architecture to decouple data reception and transmission processes, accommodating bursty or variable-rate data streams while providing reliable storage and flow control through FULL and EMPTY flags. This significantly enhances the slave's capability to manage data throughput without risk of overflow or underflow errors.

II. I²C SPECIFICATION

The I²C (Inter-Integrated Circuit) protocol is a simple process to communicate using the two wires as they are SDA and SCL and it also allows multiple devices, called masters and slaves, where each device as unique address. I²C supports different speeds Standard which comes under 100kHz, Fast (400kHz), Fast-mode Plus (1 MHz), and High-speed (3.4 MHz). It has features like clock stretching and arbitration. Clock stretching is a process for slows down the communication when device is needed. Arbitration is to prevent conflicts when multiple masters send data. I²C uses pull-up resistors to keep signals stable it is reliable and mostly used for sensors, memory chips and microcontrollers etc., It is simple design which makes perfect for small devices in IoT applications. There two main specifications about I²C.

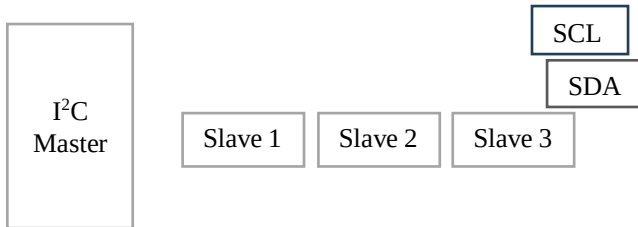
1. Two-Wire Communication

I²C is known for serial communication protocol which is done by two-wire communication. They are SDA (Serial Data Line) and SCL (Serial Clock Line). The SDA will work as data transfer between two devices, whereas SCL is a synchronizes data transmission it means the sender and receiver are aligned in timing so that the data is transmitted and received correctly.



2. Multi-Master and Multi-Slave Architecture

I²C supports multiple masters and slaves on the same bus. In this each device has a unique 7-bit or 10-bit address, allowing flexible communication. Masters initiate communication by generating start and stop conditions and also it ensures smooth operation when multiple masters attempt to communicate simultaneously.



III FPGA AND IT'S FEATURES

The Field-Programmable Gate Array(FPGA) Plays an important role in designing and testing a digital control system for an LCD display using the I²C protocol. An FPGA is like flexible digital board where we can create and test hardware design without building physical circuits. Using programming language like Verilog or VHDL, we can instruct the FPGA how could the system works. The FPGA then turn these instruction into real hardware actions. Since the FPGA can be reprogrammed, we can also try different designs, make changes, and improve them quickly. The FPGA is great for handling multiple tasks at once and can be adjusted as needed, making it perfect for testing and improving hardware ideas. This approach helps the team check if everything works correctly before making the final version. And it reduces the errors and time. FPGAs are widely used in modern digital design.

a. Reconfigurability

FPGA's can be reprogrammed for multiple times, were every one can modify and improve designs without changing the physical hardware. This process make ideal for prototyping and testing many different solutions in a efficient way.

b. Parallel Processing

This process is different method compare to previous one. Previous method works as sequential method as we know one after another but in FPGA's every task is done at same time functioning multiple tasks. It is suitable for multiple high-speed applications like signal processing and real time systems.

c. Hardware Customization

It mainly allows designers to create custom hardware configurations for specific tasks. FPGA's can be programmed for exact needs of an application, improving efficiency and also performance.

d. Reliability and Stability

FPGA's does not require complex operating systems, making them more stable and reliable. They are mainly used in mission-critical applications where consistent performance is essential, such as aerospace and defence systems.

IV FIFO BUFFER FUNCTIONALITY AND IMPORTANT

The FIFO (First-In, First-Out) buffer plays a critical role in enhancing the efficiency and reliability of the proposed I²C slave architecture. It acts as a temporary storage element that holds data between the input and output stages, allowing asynchronous data transfer between components operating at different speeds. In the context of I²C communication, where data often arrives in bursts or at irregular intervals, the FIFO buffer ensures smooth and continuous data flow. It decouples the data producer (master device) from the data consumer (slave processing logic), thereby eliminating timing dependencies and minimizing the risk of data loss. The FIFO can handle multiple data elements simultaneously, offering scalability through adjustable depth and width parameters, which can be configured according to system requirements. Furthermore, the FIFO integrates FULL and EMPTY status flags that provide essential flow control signals, preventing overflow and underflow conditions. This level of control significantly enhances the robustness of the communication process. When implemented on FPGA, the FIFO leverages efficient memory blocks (such as Block RAM), optimizing both resource utilization and power consumption. In combination with dynamic clock gating, the FIFO buffer contributes to a highly energy-efficient design, ensuring data integrity while minimizing dynamic power usage in low-power applications.

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- [1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529–551, April 1955. (references)
- [2] J. Clerk Maxwell, *A Treatise on Electricity and Magnetism*, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
- [3] I. S. Jacobs and C. P. Bean, “Fine particles, thin films and exchange anisotropy,” in *Magnetism*, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.
- [4] K. Elissa, “Title of paper if known,” unpublished.
- [5] R. Nicole, “Title of paper with only first word capitalized,” *J. Name Stand. Abbrev.*, in press.
- [6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, “Electron spectroscopy studies on magneto-optical media and plastic substrate interface,” *IEEE Transl. J. Magn. Japan*, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetics Japan, p. 301, 1982].
- [7] M. Young, *The Technical Writer’s Handbook*. Mill Valley, CA: University Science, 1989.
- [8] K. Eves and J. Valasek, “Adaptive control for singularly perturbed systems examples,” *Code Ocean*, Aug. 2023. [Online]. Available: <https://codeocean.com/capsule/4989235/tree>
- [9] D. P. Kingma and M. Welling, “Auto-encoding variational Bayes,” 2013, arXiv:1312.6114. [Online]. Available: <https://arxiv.org/abs/1312.6114>
- [10] S. Liu, “Wi-Fi Energy Detection Testbed (12MTC),” 2023, gitHub repository. [Online]. Available: <https://github.com/liustone99/Wi-Fi-Energy-Detection-Testbed-12MTC>
- [11] “Treatment episode data set: discharges (TEDS-D): concatenated, 2006 to 2009.” U.S. Department of Health and Human Services, Substance Abuse and Mental Health Services Administration, Office of Applied Studies, August, 2013, DOI:10.3886/ICPSR30122.v2

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